

Pinned group summary: $\mathrm{SL}(\mathbf{n}=2)$

Pinning-Sympy automated output

July 6, 2026

Contents

1 Basic group attributes

Group	SL (n=2)
Matrix size	2×2
Rank	1
Defining equations	<i>DefiningEquationPlaceholder</i>

1.1 Generic Lie algebra element

$$\begin{bmatrix} x_{00} & x_{01} \\ x_{10} & -x_{00} \end{bmatrix}$$

1.2 Generic torus element

$$\begin{bmatrix} t_0 & 0 \\ 0 & \frac{1}{t_0} \end{bmatrix}$$

1.3 Trivial characters

$$\begin{bmatrix} 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	A1
Irreducible	True
Reduced	True
Simply laced	True
Number of roots	2

2.1 Dynkin diagram



2.2 Cartan matrix

[2]

2.3 Roots and coroots

Root	Coroot	Simple	Sign	Multipliable
(−1, 1)	(−1, 1)	No	−	No
(1, −1)	(1, −1)	Yes	+	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ , the set of positive integral linear combinations of α, β that are also roots is denoted $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$

No pairs of roots generate positive integer linear combinations.

3 Root spaces

Root	Dimension
(−1, 1)	1
(1, −1)	1

RootSpaceTablePlaceholder

4 Root subgroups

RootSubgroupTablePlaceholder
HomomorphismDefectTablePlaceholder
HomomorphismDefectIdentitiesPlaceholder

5 Commutators

CommutatorCoefficientTablePlaceholder
CommutatorIdentitiesPlaceholder

6 Weyl group

WeylGroupPlaceholder
WeylConjugationCoefficientPlaceholder
WeylConjugationIdentitiesPlaceholder

7 Coroot torus elements (h_α)

CorootTorusElementPlaceholder